

Unit 4: Rigid Motions and Congruence

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- *Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.*
- *Clarification 2: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 3: Within the Geometry course, rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.*

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or coordinates.*
- *Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation.*
- *Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.*

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes two or more transformations.*

MA.912.GR.2.6 Apply rigid transformation to map one figure onto another to justify that the two figures are congruent.

- *Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.*

Unit 4, Lesson 1: Algebraic Descriptions for Translations and Rotations



MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.



- ☐ I can relate a translation on the coordinate plane to an algebraic notation of the transformation.
- ☐ I can relate a rotation about the origin on the coordinate plane to an algebraic notation of the transformation.



A career as a pharmacist is just one of many jobs in the health care field. Pharmacists need an understanding of both science and mathematics. Pharmacists also need to work with patients on their prescriptions to ensure the medication is working as it should.



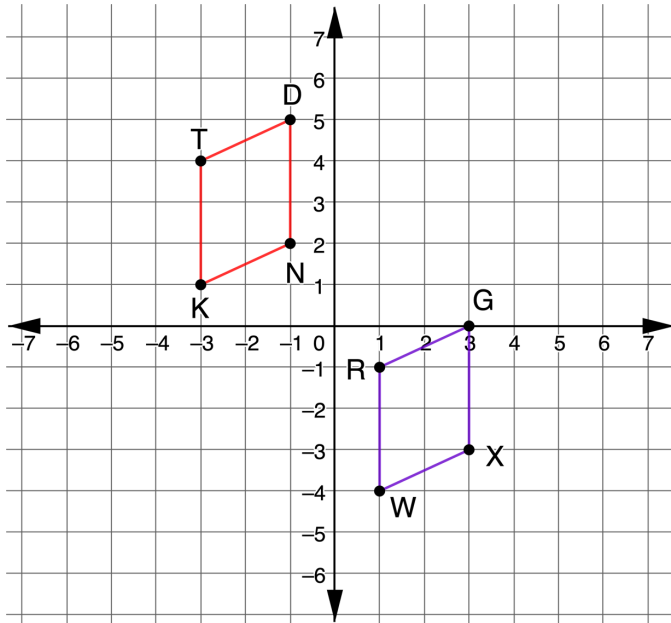
1. What is one thing you learned about pharmacists that you did not know?
2. How does paying attention to detail help you to learn mathematics?





4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

1. Parallelograms $DNKT$ and $GXWR$ are shown on the coordinate grid.



- a. Describe the transformation if parallelogram $DNKT$ is the preimage.
- b. Describe the transformation if parallelogram $GXWR$ is the preimage.

- c. Complete the table by writing the x -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the x -value for the vertices of parallelogram $DNKT$ to get the x -value for the vertices of parallelogram $GXWR$.

Vertex	x	Vertex	x	Description
D		G		☐ Add ☐ Subtract _____
N		X		☐ Add ☐ Subtract _____
K		W		☐ Add ☐ Subtract _____
T		R		☐ Add ☐ Subtract _____

- d. Complete the table by writing the y -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the y -value for the vertices of parallelogram $DNKT$ to get the y -value for the vertices of parallelogram GXR .

Vertex	y	Vertex	y	Description
D		G		☐ Add ☐ Subtract
N		X		☐ Add ☐ Subtract
K		W		☐ Add ☐ Subtract
T		R		☐ Add ☐ Subtract

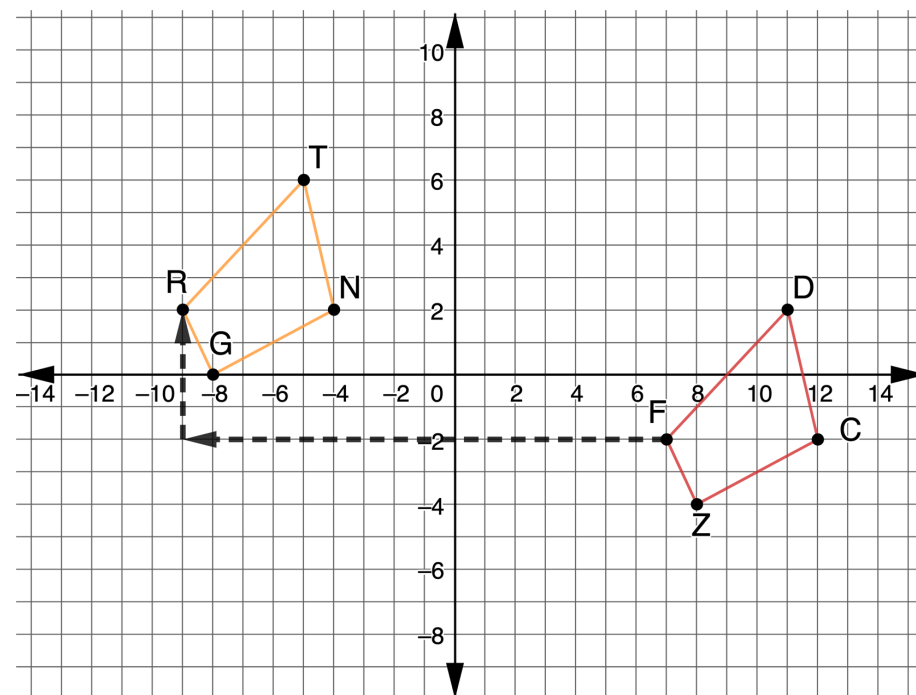
Transformations can be described using words, as well as using a notation.

- e. Use the tables to complete the algebraic description of the transformation if parallelogram $DNKT$ is the preimage.

$$(x, y) \rightarrow (x \boxed{}, y \boxed{})$$

- f. Discuss with your partner how the algebraic description of the transformation would change if parallelogram GXR is the preimage.

2. Quadrilaterals $GRTN$ and $ZFDC$ are shown on the coordinate grid. A mapping of vertex F is shown.

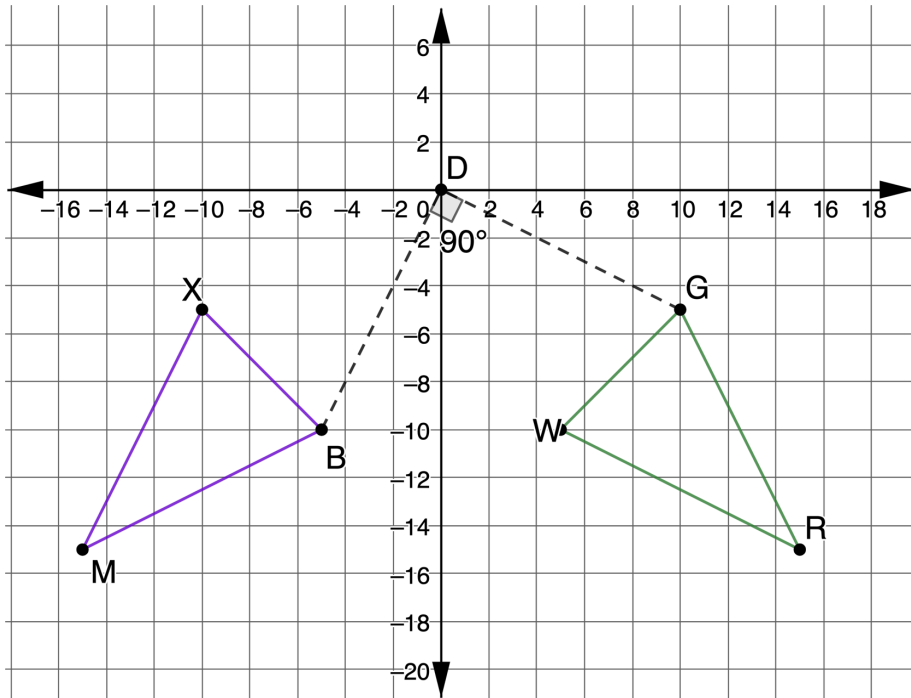


- a. Map the transformation for the other vertices of quadrilateral $ZFDC$.
- b. Discuss with your partner how the algebraic description of the transformation, $(x, y) \rightarrow (x - 16, y + 4)$, is revealed in the mapping.



4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

1. Triangles BXM and GWR are shown on the coordinate grid, where $\triangle GWR$ is the preimage. A mapping of vertex G is shown.



- a. Map the transformation for the other vertices of triangle GWR .
- b. Describe the transformation if triangle GWR is the preimage.

- c. Discuss with your partner how to use the table of values to describe the relationship of the corresponding vertices using an algebraic description.

Triangle GWR

Vertex	x	y
G	10	-5
W	5	-10
R	15	-15

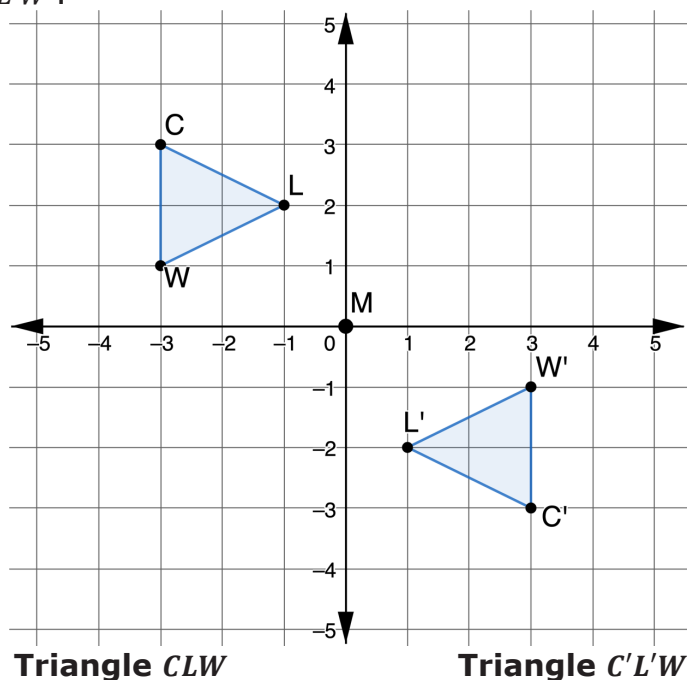
Triangle BXM

Vertex	x	y
B	-5	-10
X	-10	-5
M	-15	-15

- d. Write a conjecture that you and your partner think will work for the algebraic description of the rotation.

$$(x, y) \rightarrow (\text{_____, } \text{_____})$$

2. Triangles CLW and $C'L'W'$ are shown on the coordinate grid. Triangle CLW was rotated 180° about M to create $\Delta C'L'W'$.



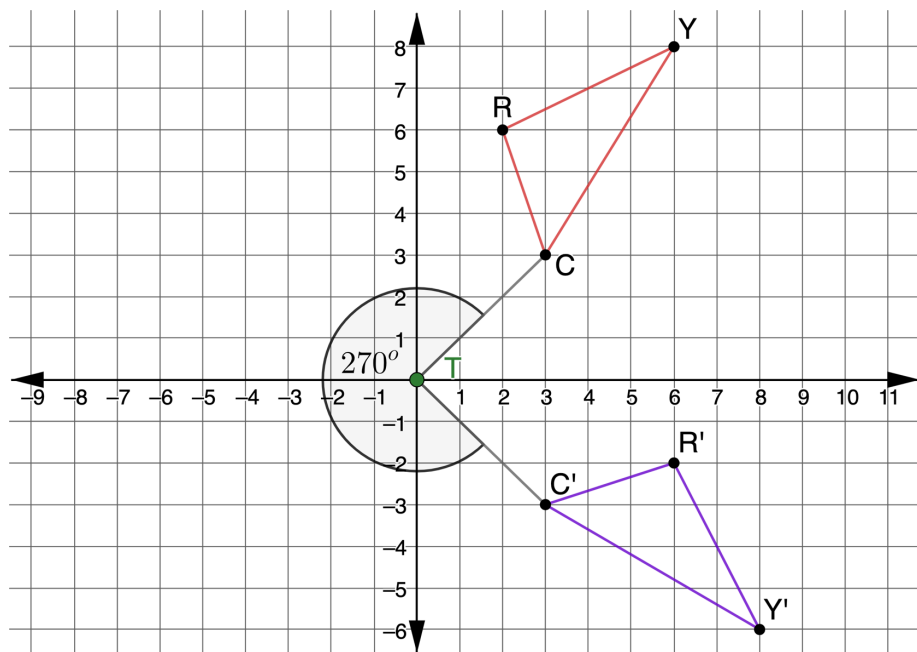
Vertex	x	y
C	-3	3
L	-1	2
W	-3	1

Vertex	x	y
C'	3	-3
L'	1	-2
W'	3	-1

- a. It was stated that ΔCLW was rotated 180° about M to create $\Delta C'L'W'$. Was that rotation clockwise or counterclockwise?

- b. Discuss with your partner how your previous conjecture works with rotating ΔCLW 180° about M . If necessary, write a new conjecture that works for rotations of 180° about the origin. Include the direction of the rotation determined.
- c. Assuming $\Delta C'L'W'$ was the preimage and is rotated 180° about the origin, test whether your previous or new conjecture works to describe this rotation. Include the direction of rotation.

3. Triangles YCR and $Y'C'R'$ are shown on the coordinate grid. $\triangle YCR$ was rotated 270° counterclockwise about T to create $\triangle Y'C'R'$.



Triangle YCR

Vertex	x	y
Y	6	8
C	3	3
R	2	6

Triangle $Y'C'R'$

Vertex	x	y
Y'	8	-6
C'	3	-3
R'	6	-2

- a. Discuss with your partner how the ordered pairs for $\triangle YCR$ and $\triangle Y'C'R'$ could be used to write a conjecture that works for rotations of 270° counterclockwise about the origin.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

- b. Test your conjecture for rotations of 270° counterclockwise by looking at $\triangle BXM$ and $\triangle GWR$ from question 1. Assume $\triangle GWR$ is the preimage and was rotated 270° counterclockwise about the origin. If necessary, adjust your conjecture.
- c. Discuss with your partner why the same conjecture should work for a rotation of 270° counterclockwise and 90° clockwise about the origin.



4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Translations	
Right	Left
Add to the x -coordinate $(x, y) \rightarrow (x + \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	Subtract from the x -coordinate $(x, y) \rightarrow (x - \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
Up	Down
Add to the y -coordinate $(x, y) \rightarrow (\underline{\hspace{1cm}}, y + \underline{\hspace{1cm}})$	Subtract from the y -coordinate $(x, y) \rightarrow (\underline{\hspace{1cm}}, y - \underline{\hspace{1cm}})$

Rotations clockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
Rotations counterclockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$	$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

1. What do you notice about the algebraic descriptions for rotations?



4.1.5 Wrap-Up: Cool Down

1. A student asks you for help with algebraic descriptions for rotations because they are stuck on how to do it. Describe the process using what you learned in this lesson, highlighting any potential misconceptions they may have.

Unit 4, Lesson 2: Algebraic Descriptions for Reflections



Benchmark

MA.912.GR.2.1 Given a pre-image and image, describe the transformation and represent the transformation algebraically using coordinates.



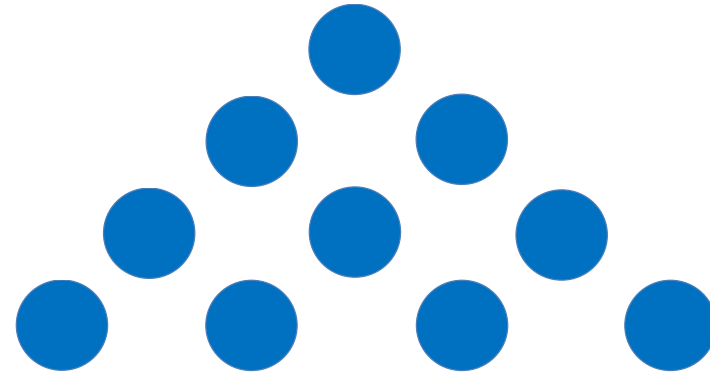
Learning Target

- ☐ I can relate a reflection on the coordinate plane to an algebraic notation of the transformation.



4.2.1 Warm-Up: Moving Circles

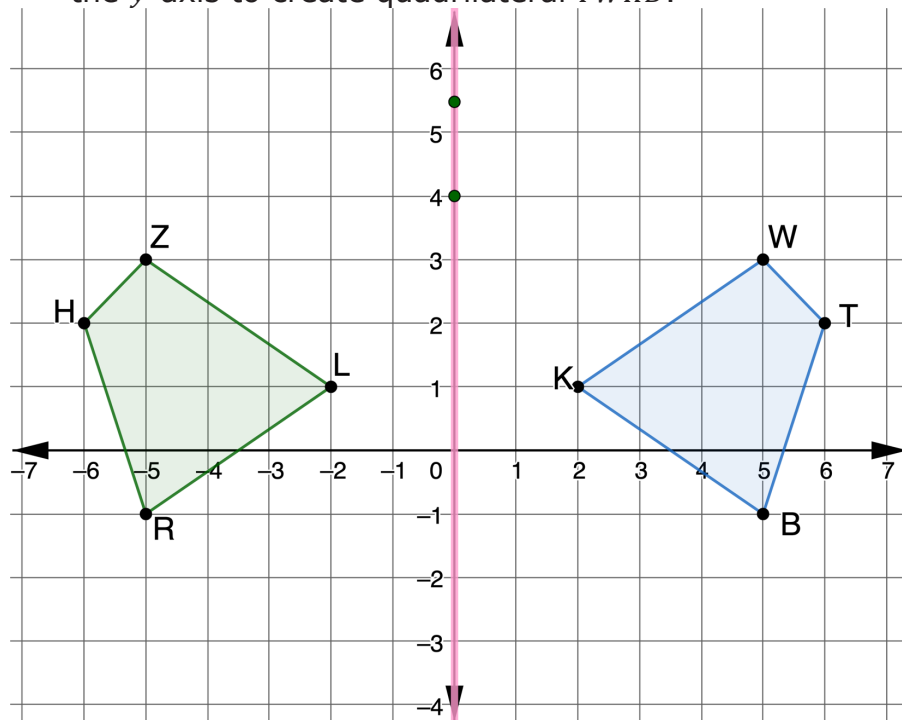
1. Move any three circles so that the pyramid points down.





4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

1. Quadrilaterals $HZLR$ and $TWKB$ are shown on the coordinate grid. Quadrilateral $HZLR$ was reflected across the y -axis to create quadrilateral $TWKB$.



- Draw a line segment between each set of corresponding vertices.
- Describe how the line segments between the vertices are related to the y -axis.

- c. Complete the table.

	H to T	Z to W	L to K	R to B
Distance				
Ordered Pair of the Midpoint				

- d. What do you notice about the midpoints?

- e. The table shows the coordinates of each vertex.

Vertex	x	y	Vertex	x	y
H	-6	2	T	6	2
Z	-5	3	W	5	3
R	-2	1	K	2	1
L	-5	-1	B	5	-1

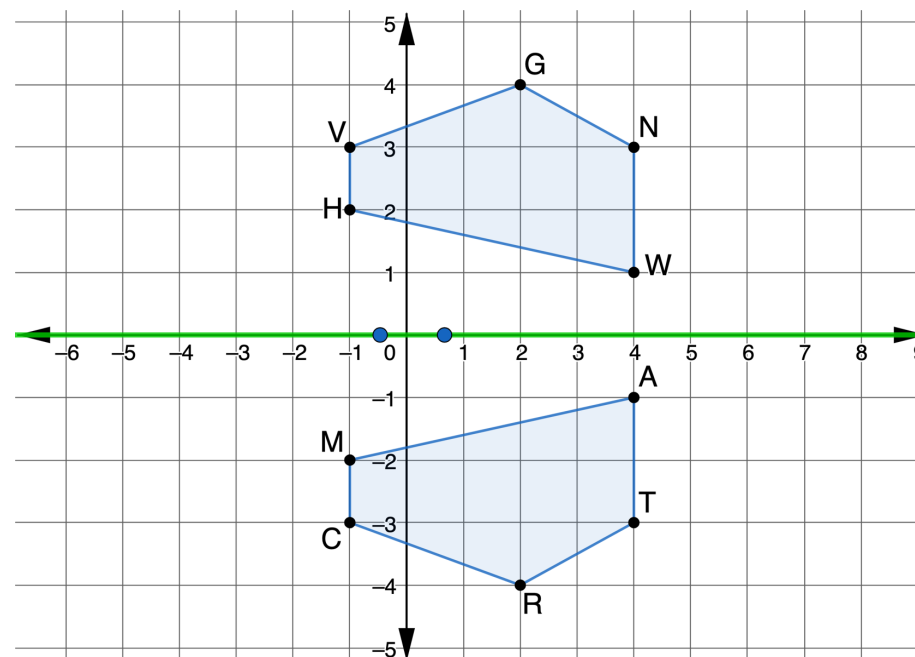
Work with your partner to describe how the coordinates of the pre-image, quadrilateral $HZLR$, are related to the coordinates of the image, quadrilateral $TWKB$.

- f. Write a conjecture that you and your partner think will work for the algebraic description of a reflection.

$(x, y) \rightarrow (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

- g. Discuss with your partner if the algebraic description will work for any reflection. Summarize your conclusion.

2. Polygons $MCRTA$ and $HVGNW$ are shown on the coordinate grid. Polygon $MCRTA$ was reflected across the x -axis to create polygon $HVGNW$.



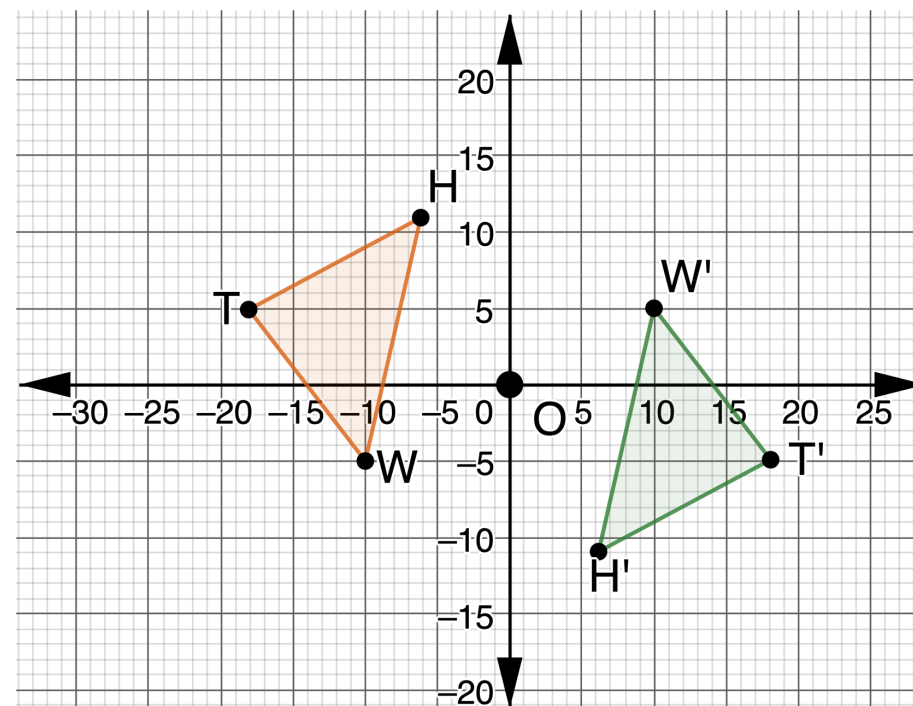
- a. Describe the relationship that the line segments between each set of corresponding vertices have with the x -axis.

- b. Wendall and Samah were working together on writing an algebraic description for reflections across the x -axis. Each student's algebraic description and explanation of their thinking is shown in the table.

Wendall	Samah
$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x, -y)$
Because the polygon flipped over the x -axis, the x becomes its opposite.	Because above the x -axis all y -values are positive and below the x -axis all y -values are negative, the y becomes its opposite.

Discuss with your partner which, if any, algebraic description is correct. Use at least two ordered pairs from polygons $MCRTA$ and $HVGNW$ to defend your choice.

3. Triangles HWT and $H'W'T'$ are shown on the coordinate grid. Triangle HWT was reflected across the origin to create $\triangle H'W'T'$.

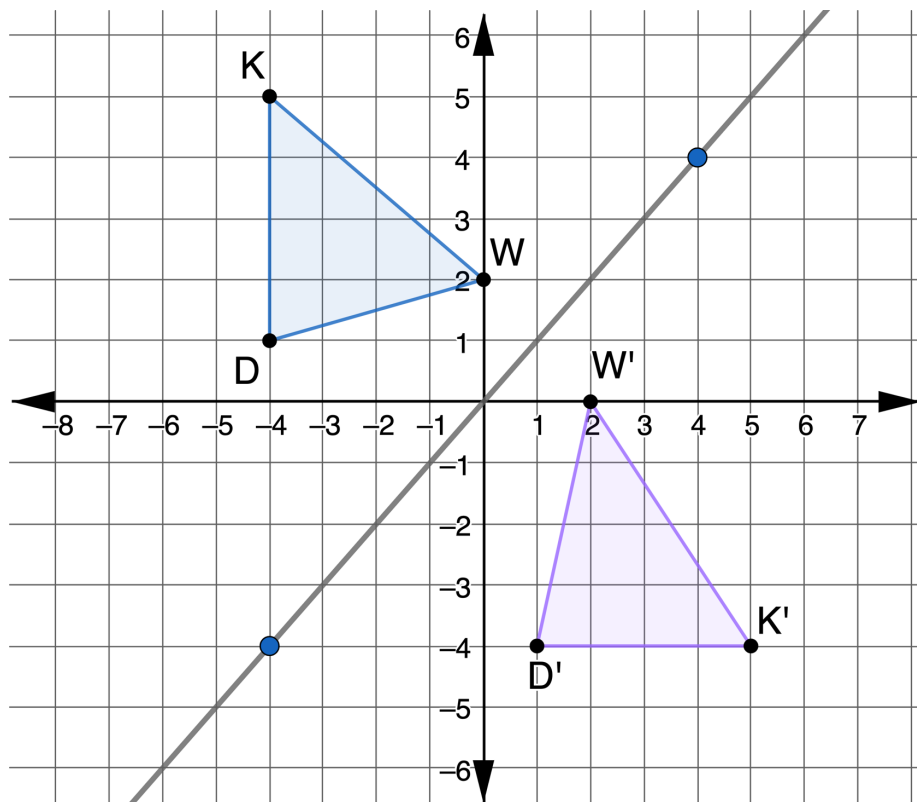


- a. Discuss with your partner how to write the algebraic description for a reflection across the origin.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

- b. Re-examine the algebraic descriptions of rotations. Which rotation is similar to a reflection across the origin?

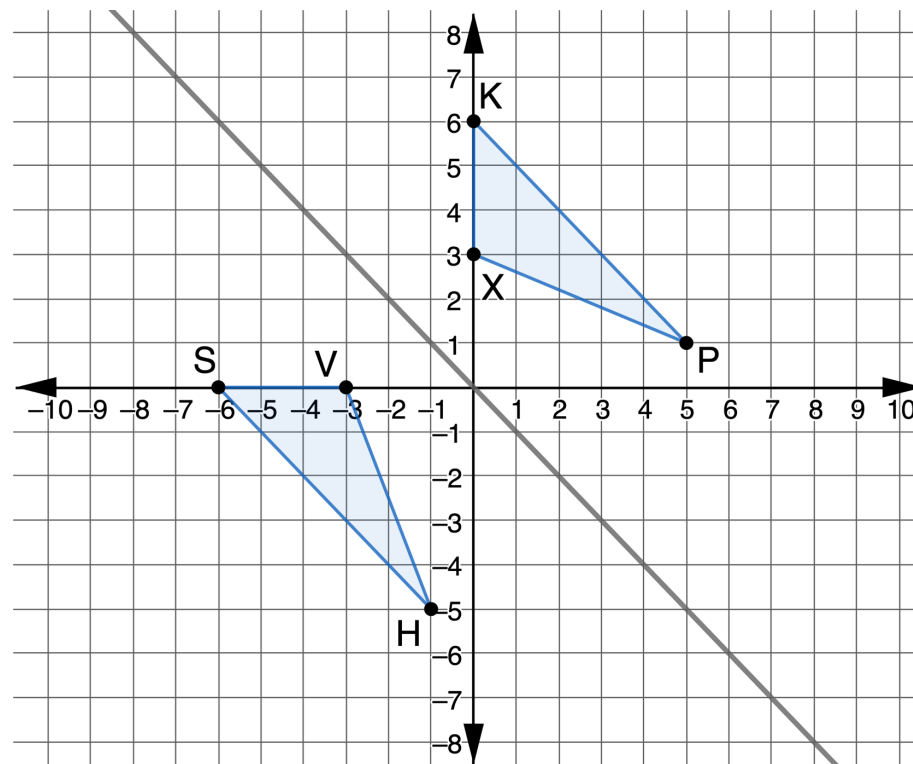
4. Triangles KWD and $K'W'D'$ are shown on the coordinate grid. Triangle KWD was reflected across the line $y = x$ to create $\triangle K'W'D'$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = x$.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

5. Triangles HSV and PKX are shown on the coordinate grid. Triangle HSV was reflected across the line $y = -x$ to create $\triangle PKX$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = -x$.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$



4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Reflections	
Across the x -axis	Across the y -axis
The ____-coordinate changes to its ____. $(x, y) \rightarrow (_, _)$	The ____-coordinate changes to its ____. $(x, y) \rightarrow (_, _)$
Across $y = x$	Across $y = -x$
The coordinates ____. $(x, y) \rightarrow (_, _)$	The coordinates ____ and become ____. $(x, y) \rightarrow (_, _)$



4.2.4 Wrap-Up: Cool Down

A reflection can be thought of as folding or "flipping" an object over the line of reflection. A squish painting is a fun application of a line of reflection.

To make a squish painting, a large sheet of construction paper is folded in half then opened up. Paint is dotted along one side of the fold. The paper is folded, and the folded paper is rubbed, pounded, and squished to move the paint around. The paper is then opened up to reveal the squish painting.



1. How does a squish painting show that in a reflection, each point of the original figure and its image are the same distance away from the line of reflection?

Unit 4, Lesson 3: Sequences of Transformations



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.



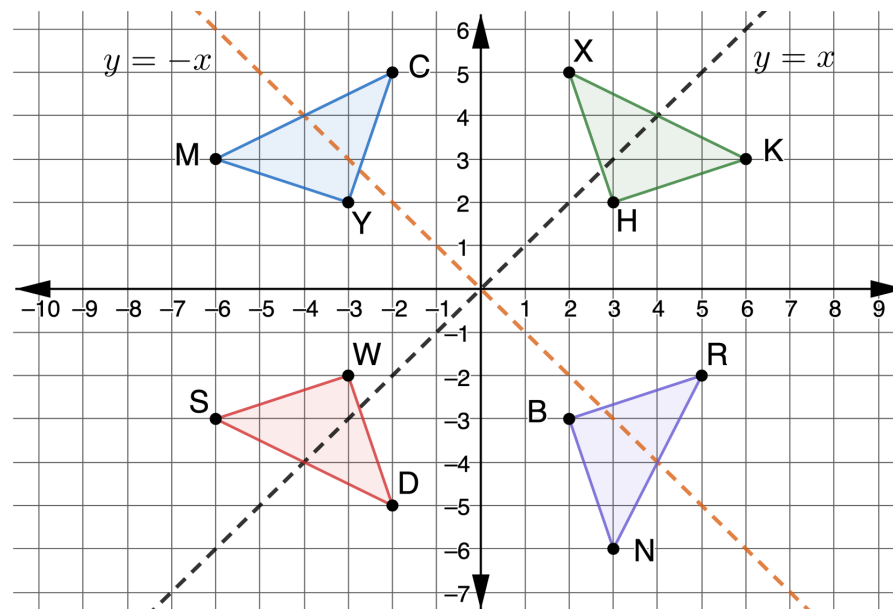
Learning Targets

- ☐ I can identify transformations that preserve distance.
- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.



4.3.1 Warm-Up: Bell Work

Triangles MCY , KXH , NRB , and SWD are shown on the coordinate grid.



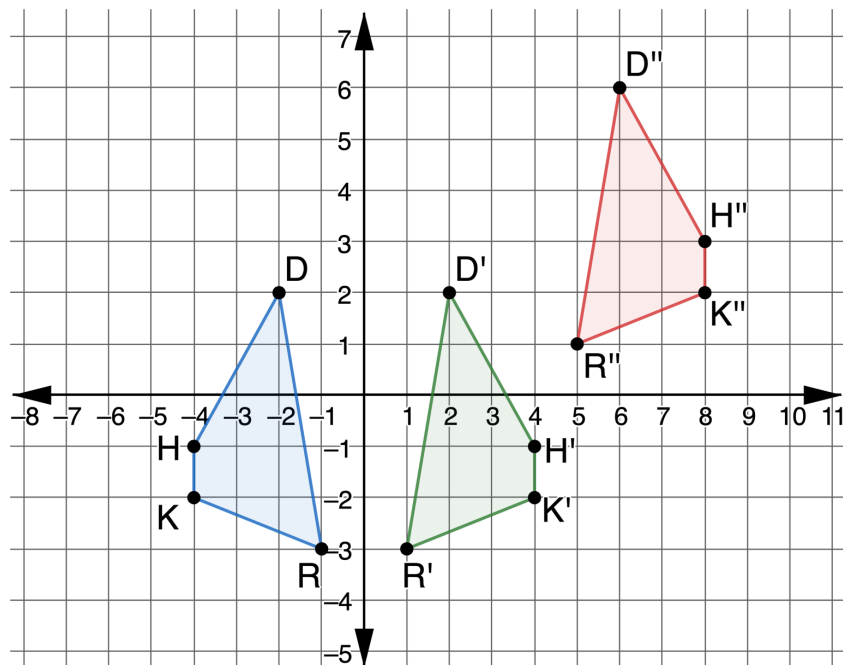
Complete each statement.

- When $\triangle MCY$ is reflected across _____, it will map onto $\triangle KXH$.
- When $\triangle NRB$ is reflected across _____, it will map onto $\triangle$ _____.
- When $\triangle SWD$ is reflected across the x -axis, it will map onto $\triangle$ _____.



4.3.2 Guided Instruction: Sequence of Transformations

1. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



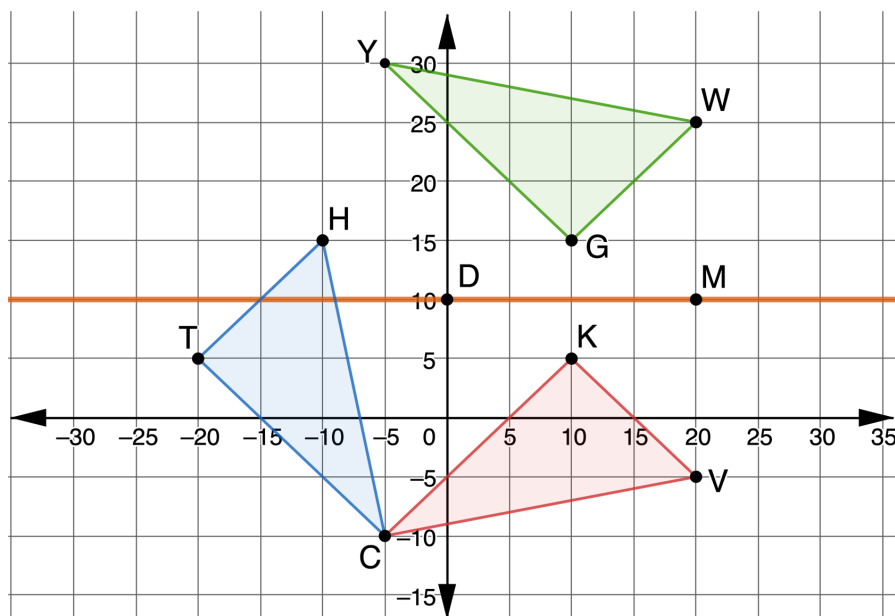
- a. On the graph, show the mappings of the vertices of the first transformation.
- b. On the graph, show the mappings of the vertices of the second transformation.

- c. Complete the table.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description		
Algebraic Description		

- d. Explain how to know if a sequence of transformations preserves distance.

2. Triangles HTC , VKC , and WGY are shown on the coordinate grid. Triangle HTC is the pre-image. Triangle VKC is the result of the first transformation, and $\triangle WGY$ is the result of the second transformation.



a. Describe the first transformation.

b. Use patty paper to show that the transformation preserved distance.

c. Describe the second transformation.

d. Use patty paper to show that the second transformation preserved distance.

e. Work with your partner to complete the statement.

Because the ☐ center of rotation ☐ line of reflection of ☐ $\triangle HTC$ ☐ $\triangle VKC$

is ☐ $(0, 0)$, ☐ $(-5, -10)$, ☐ $y = 0$, ☐ $y = 10$, an algebraic description

☐ can ☐ cannot be used.

f. When Judeline and Carmen completed the statement above, they had different correct answers. Write the second correct answer.

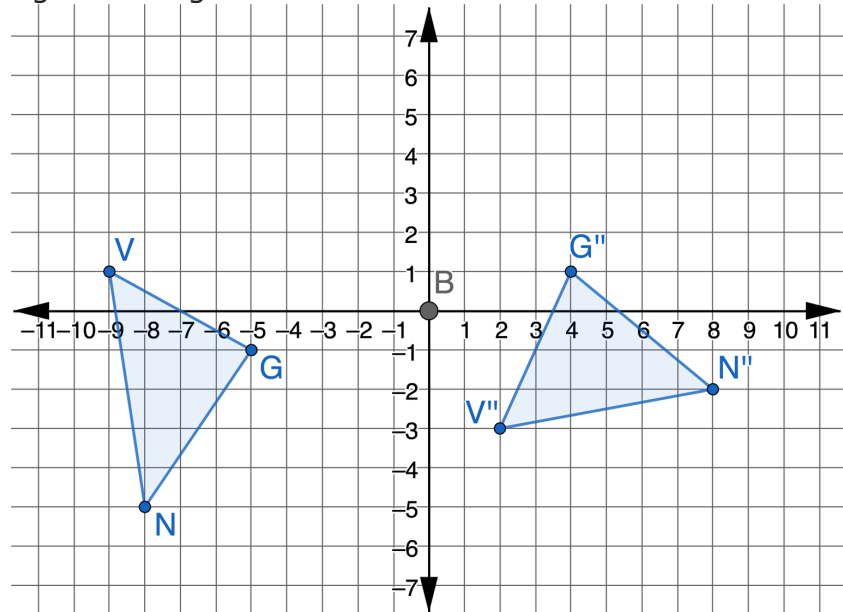


To use an algebraic description of a rotation the center of rotation must be _____.



To use an algebraic description of a reflection the line of reflection must be _____, _____, or _____.

3. Triangles NVG and $N''V''G''$ are shown on the coordinate grid. Triangle NVG was transformed to create $\Delta N''V''G''$.



- a. Explain which features of the two triangles can be used to determine that the transformations were rigid motions.
- b. Adele told Kiriam that she knows one of the rigid motions is a rotation. Discuss with your partner which clues in the location of the vertices Adele used to draw her conclusion.

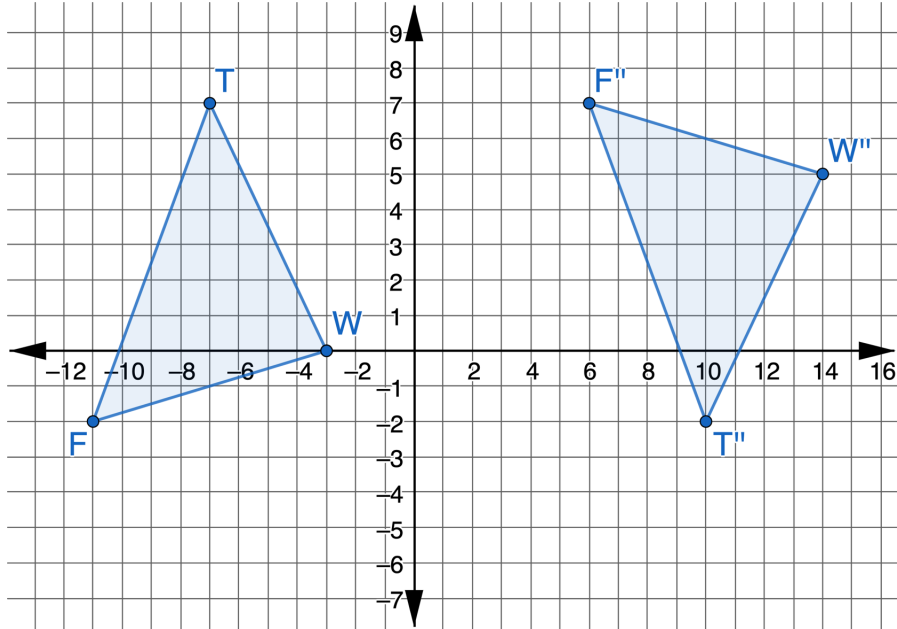
- c. Work with your partner to determine a sequence of transformations that includes a rotation about the point B so that when applied, ΔNVG will map onto $\Delta N''V''G''$. Complete the table.

Transformation	Mapping	Algebraic Description
	$\Delta NVG \rightarrow \Delta N'V'G'$	
	$\Delta N'V'G' \rightarrow \Delta N''V''G''$	



4.3.3 Guided Practice: Sequence of Transformations

- Triangles TWF and $T''W''F''$ are shown on the coordinate plane.



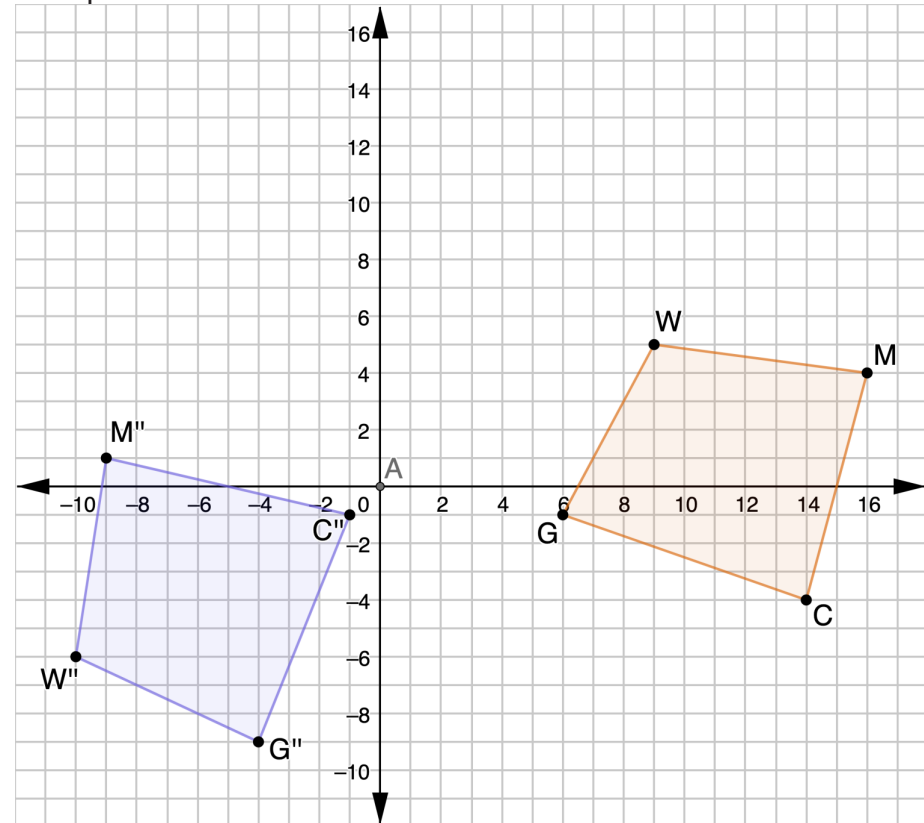
Determine a sequence of transformations that will map $\triangle TWF$ onto $\triangle T''W''F''$. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle TWF \rightarrow \triangle T'W'F'$	
	$\triangle T'W'F' \rightarrow \triangle T''W''F''$	



4.3.4 Your Turn!

- Quadrilateral $MCGW$ was transformed to create quadrilateral $M''C''G''W''$.



- Describe the sequence of transformations that will map quadrilateral $MCGW$ onto quadrilateral $M''C''G''W''$.
- Write the sequence of transformations using algebraic descriptions.



4.3.5 Wrap-Up: Reflection

1. Check the box that describes your understanding of each learning target.

- ☐ I can identify transformations that preserve distance and I am ready for the next lesson.

- ☐ I need some help identifying transformations that preserve distance. One area in which I need help is

_____.

I need more support in knowing how to identify transformations that preserve distance. I am

- ☐ concerned I will not be ready to move on. One area in which I need help is

_____.

- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.

- ☐ I need some help specifying a sequence of transformation that will map a given figure onto another congruent figure. One area in which I need help is

_____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

_____.



Unit 4, Lesson 4: Written Descriptions of a Sequence of Transformations



Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.



Learning Targets

- ☐ I can draw an image given a pre-image and a written description of a sequence of transformations on a coordinate plane.
- ☐ I can describe a transformation on the coordinate plane using written descriptions.



4.4.1 Warm-Up: Notice and Wonder

A kaleidoscope is an instrument that contains loose bits of colored material, usually glass, between two flat plates and two plane mirrors. The plates can be moved so the colored material changes position. The mirrors then reflect the material in an endless variety of patterns.

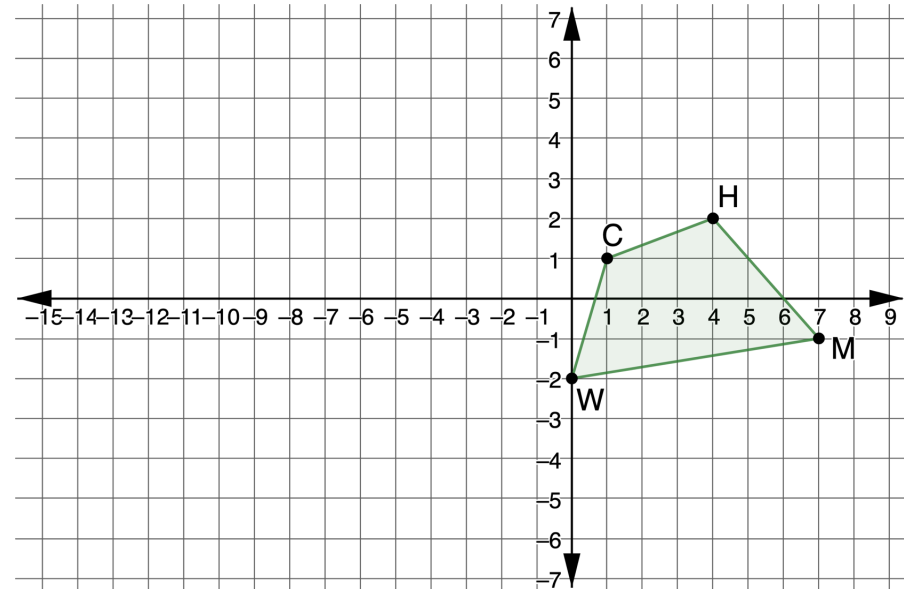


1. What do you notice about the pattern from the kaleidoscope in the picture?
2. What do you wonder about the pattern?



4.4.2 Collaboration: Drawing the Image

1. Quadrilateral $CHMW$ is shown on the coordinate grid.

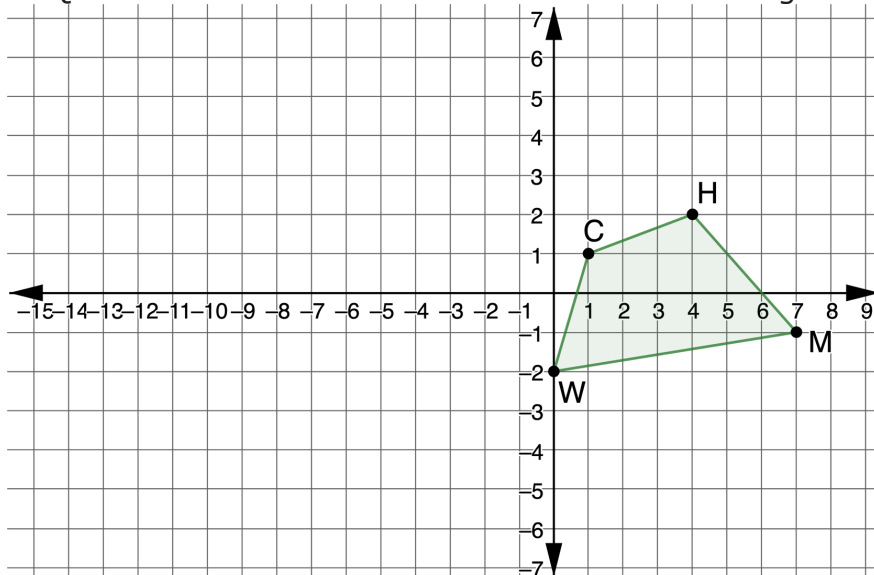


- a. Draw the quadrilateral that is the result of a reflection across the y -axis and then translated left 3 and up 1. Label the new quadrilateral $BVXN$.
- b. Check your graph with your partner. Make any necessary adjustments.

A **sequence of transformations**, or a composition of transformations, is when two or more transformations are combined.

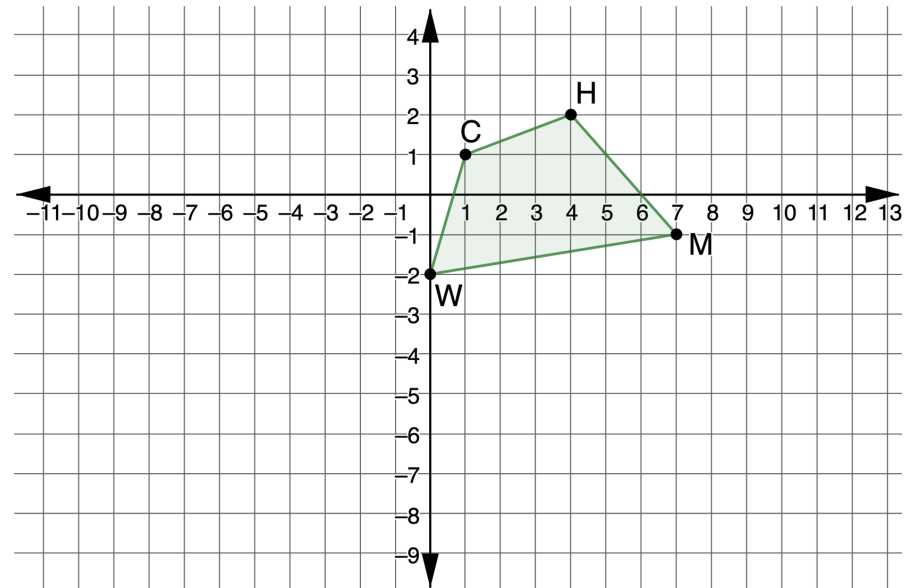
- c. Work with your partner to explain why quadrilateral $BVXN$ is a result of a sequence of transformations.

2. Quadrilateral $CHMW$ is shown on the coordinate grid.



- Draw the quadrilateral that is the result of a rotation of 180° counterclockwise about the origin and then reflected across the x -axis. Label the new quadrilateral $KDPL$.
- Check your graph with your partner. Make any necessary adjustments.

3. Quadrilateral $CHMW$ is shown on the coordinate grid.



- Draw the quadrilateral that is the result of a rotation of 90° clockwise about the point W and then translated left 2 and up 5. Label the new quadrilateral $RYFG$.
- Check your graph with your partner. Make any necessary adjustments.

4. For each sequence of transformations, summarize your mistakes and what you learned from them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- a. A reflection across the y -axis and then translated left 3 and up 1
- b. A rotation of 180° counterclockwise about the origin and then reflected across the x -axis
- c. A rotation of 90° clockwise about the point W and then translated left 2 and up 5



4.4.3 Guided Instruction: Using Coordinates

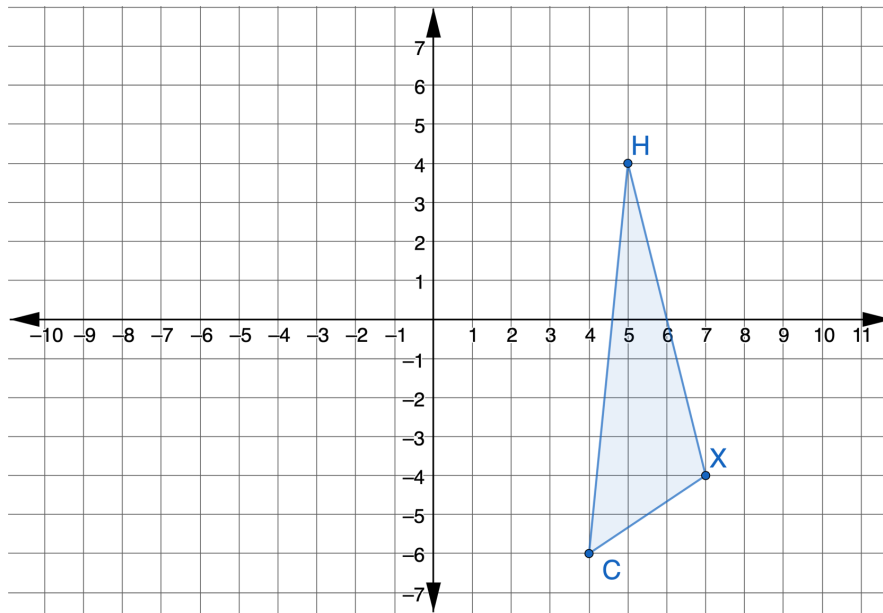
The Effect of Transformations on Coordinates

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its _____.
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become their opposite.

1. Triangle CHX is shown on the coordinate grid.



Triangle CHX

Vertex	x	y
C	4	-6
H	5	4
X	7	-4

- a. Reflect $\triangle CHX$ across the y -axis then rotate 90° clockwise about the origin. Show both $\triangle C'H'X'$ and $\triangle C''H''X''$.
- b. Complete the table of values for each vertex.

Triangle $C'H'X'$

Vertex	x	y
C'		
H'		
X'		

Triangle $C''H''X''$

Vertex	x	y
C''		
H''		
X''		

- c. Discuss with your partner how the tables show the effect of transformations on coordinates.

One strategy to apply transformations is to use tables of ordered pairs and then to plot the ordered pairs.

2. The table of values shows the coordinates of the vertices of quadrilateral $WBKT$.

Quadrilateral $WBKT$

Vertex	x	y
W	-17	-5
B	-11	-5
K	-4	-11
T	-14	-16

- a. Complete the table of values for the given sequence of transformations of quadrilateral $WBKT$.

Rotate quadrilateral $WBKT$ 180° counterclockwise about the origin, then translate it right 7 down 4.

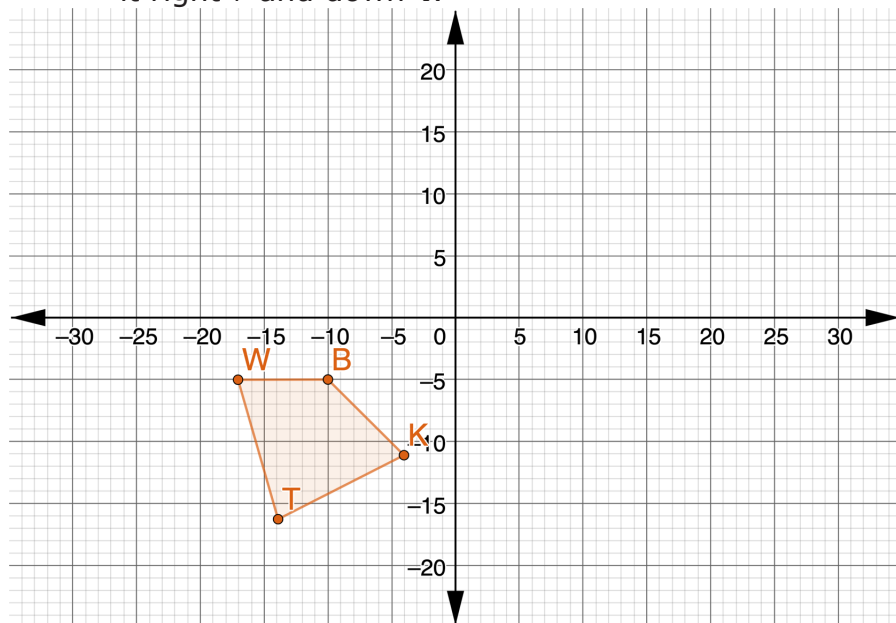
Quadrilateral $W'B'K'T'$

Vertex	x	y
W'		
B'		
K'		
T'		

Quadrilateral $W''B''K''T''$

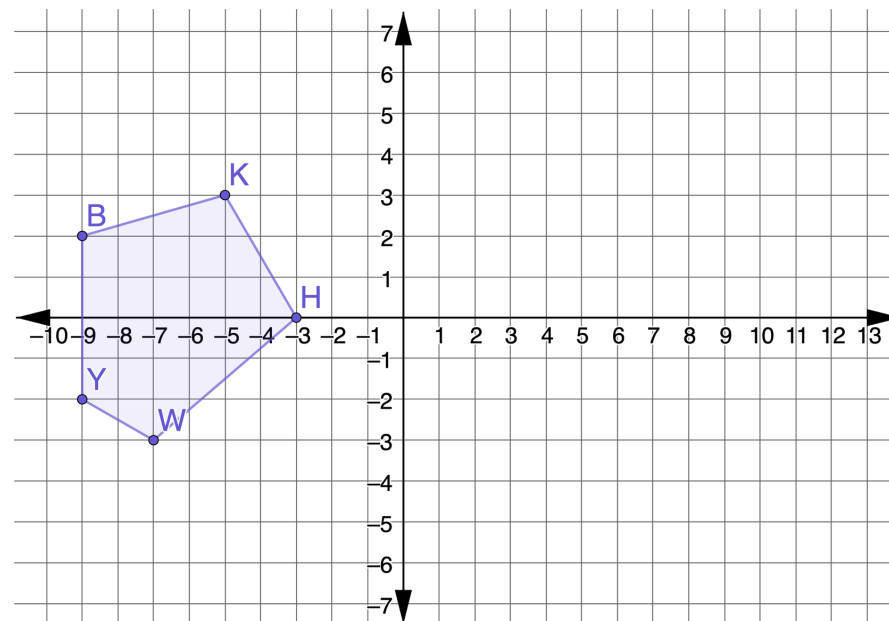
Vertex	x	y
W''		
B''		
K''		
T''		

- b. Quadrilateral $WBKT$ is shown on the coordinate grid. Show the sequence of transformations on the coordinate grid by rotating quadrilateral $WBKT$ 180° counterclockwise about the origin, then translating it right 7 and down 4.



4.4.4 Your Turn!

1. Polygon $BKHWY$ is shown on the coordinate grid.

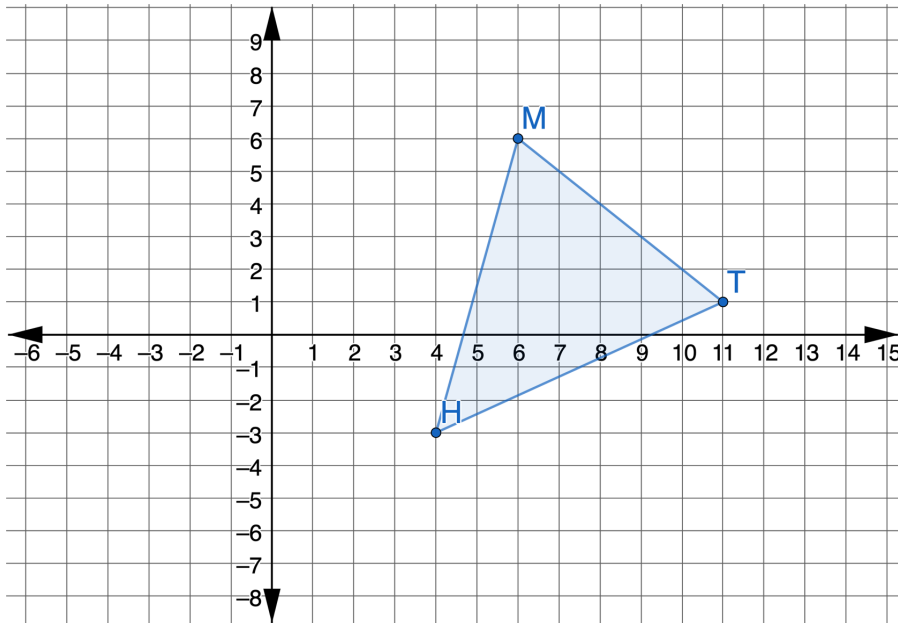


On the coordinate grid, translate polygon $BKHWY$ right 5 and down 3. Then, reflect the result across the line $x = 4$. Label the resulting polygon $B''K''H''W''Y''$.



4.4.5 Wrap-Up: Ticket Out the Door

1. Triangle HMT is shown on the coordinate grid.



Draw $\triangle BWF$, which is the result of translating $\triangle HMT$ left 3 and up 3, then rotating the result 90° clockwise about the origin.

Unit 4, Lesson 5: Algebraic Descriptions of a Sequence of Transformations



Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.



Learning Targets

- ☐ I can draw an image given a pre-image and an algebraic description of a sequence of transformations on a coordinate plane.
- ☐ I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.



4.5.1 Warm-Up: Algebraic Descriptions of Transformations

1. Use your notes to complete the algebraic descriptions in the table.

The Effect of Transformations on Coordinates

a.

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$

b.

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$

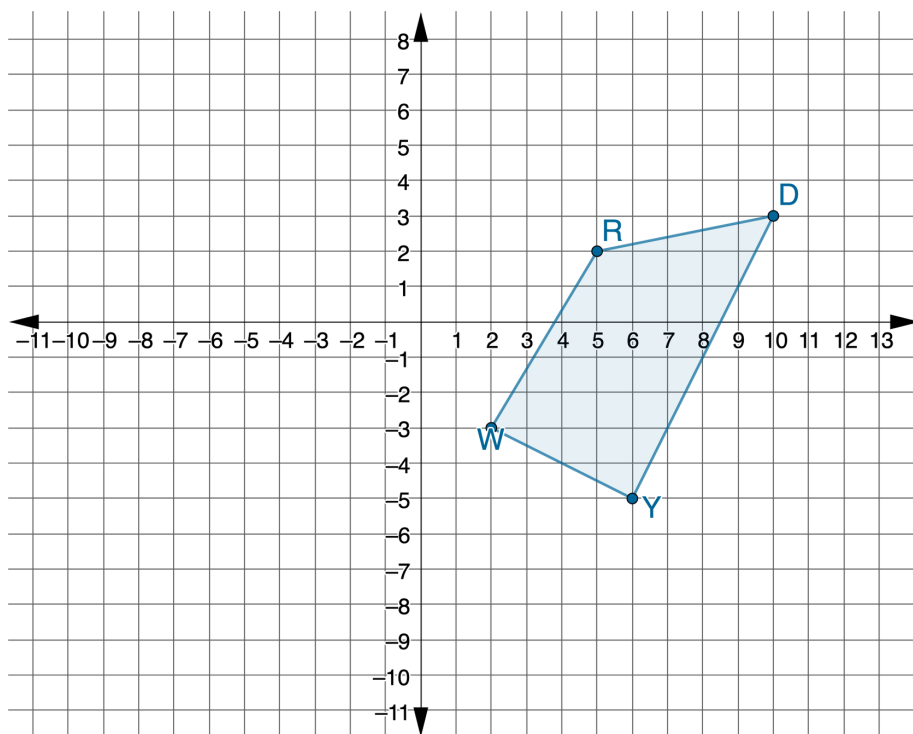
c.

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its opposite.
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become the opposite.
$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$	$(x, y) \rightarrow (\underline{\quad}, \underline{\quad})$



4.5.2 Guided Instruction: Applying Algebraic Descriptions

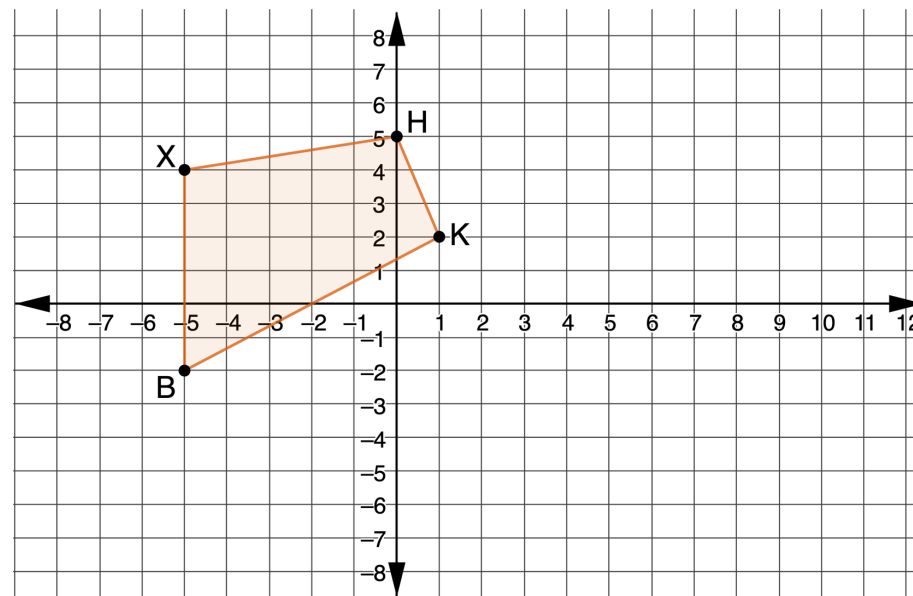
1. Quadrilateral $RDYW$ is shown on the coordinate grid.



Apply the sequence of transformations to quadrilateral $RDYW$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (-x, y)$ Label as $R'D'Y'W'$.
- $(x, y) \rightarrow (-y, x)$ Label as $R''D''Y''W''$.

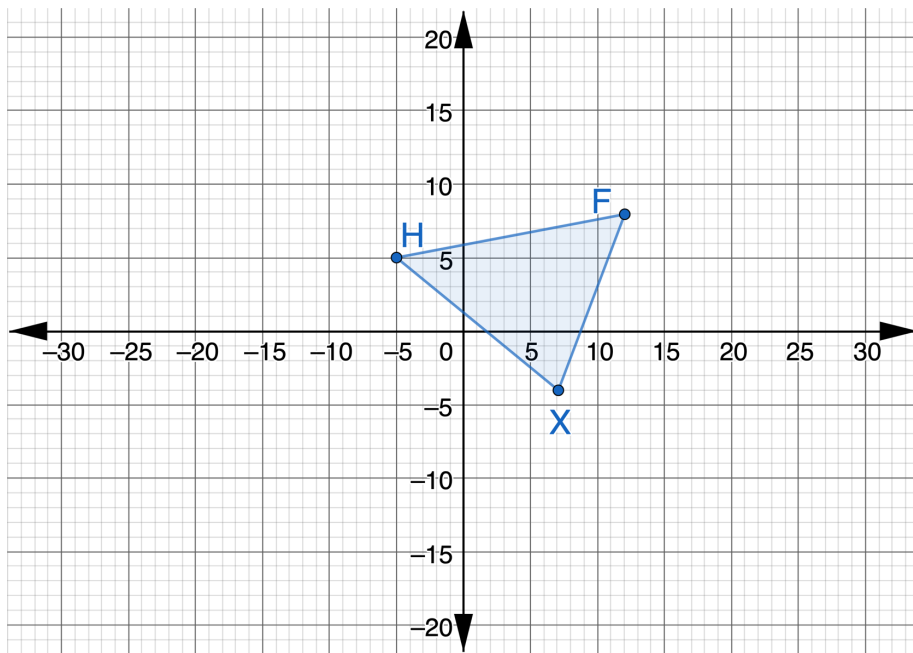
2. Quadrilateral $BXHK$ is shown on the coordinate grid.



Apply the series of transformations to quadrilateral $BXHK$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (x + 10, y - 4)$ Label as $B'X'H'K'$.
- $(x, y) \rightarrow (x, -y)$ Label as $B''X''H''K''$.

3. Triangle HFX is shown on the coordinate grid.



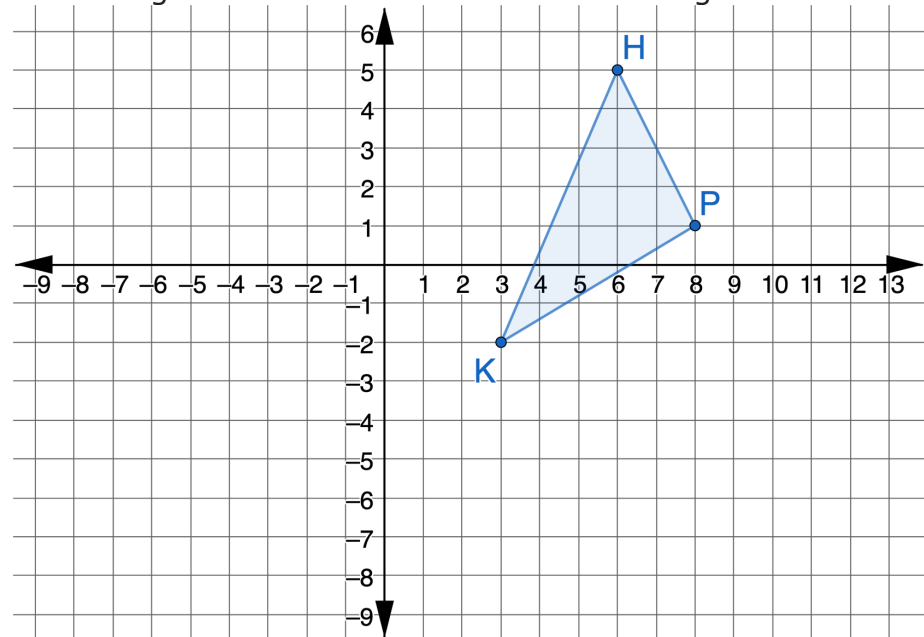
Perform the sequence of transformations on $\triangle HFX$. Assume all rotations are centered at the origin. Label the result of all three transformations as $\triangle TMR$.

- $(x, y) \rightarrow (x - 22, y + 9)$
- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-x, -y)$



4.5.3 Guided Practice: Applying Algebraic Descriptions

1. Triangle PKH is shown on the coordinate grid.

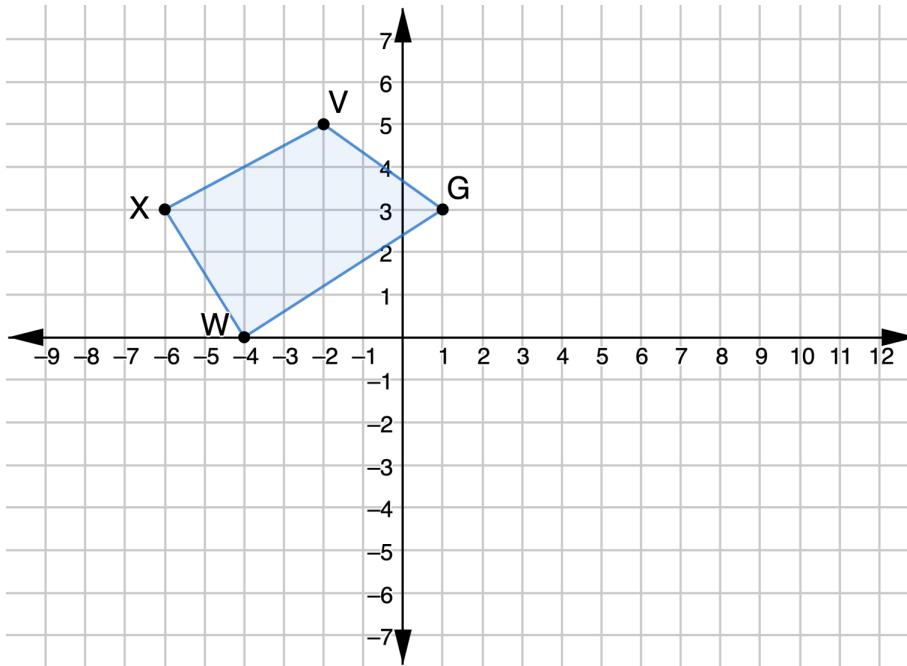


a. Perform the sequence of transformations on $\triangle PKH$. Assume all rotations are centered at the origin. Label the result of transformations as $\triangle YTM$.

- $(x, y) \rightarrow (x, -y)$
- $(x, y) \rightarrow (x - 9, y - 3)$

b. Check your answer with your partner.

2. Quadrilateral $XVGW$ is shown on the coordinate grid.



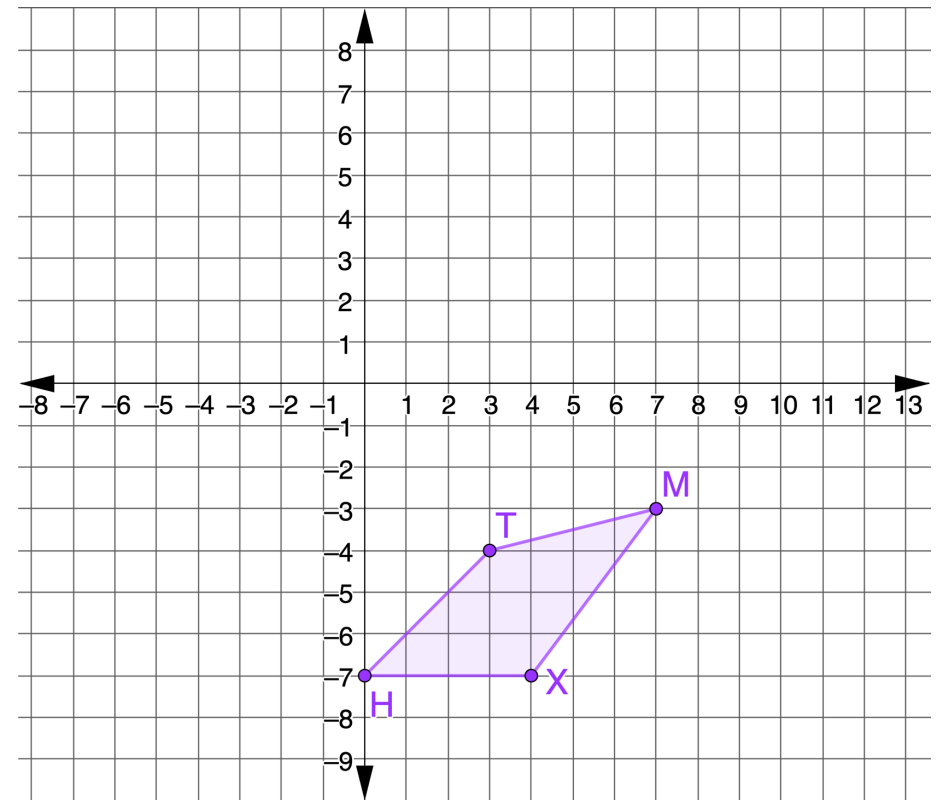
Perform the series of transformations on quadrilateral $XVGW$. Assume all rotations are centered at the origin. Label the result of transformations as quadrilateral $BHMK$.

- $(x, y) \rightarrow (-x, -y)$
- $(x, y) \rightarrow (y, -x)$
- $(x, y) \rightarrow (x + 4, y - 1)$



4.5.4 Your Turn!

1. Quadrilateral $XHTM$ is shown on the coordinate grid.



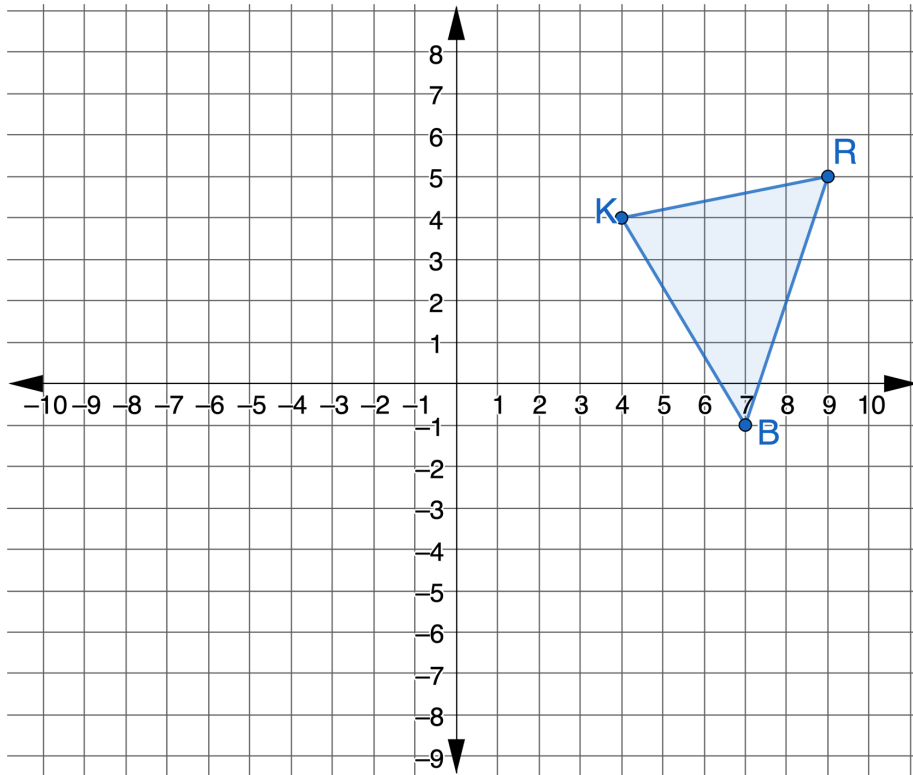
Perform the sequence of transformations on quadrilateral $XHTM$. Assume all rotations are centered at the origin. Label the result of the transformations as quadrilateral $WQPZ$.

- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-y, x)$
- $(x, y) \rightarrow (x + 4, y - 1)$



4.5.5 Wrap-Up: Ticket Out the Door

1. Triangle KRB is shown on the coordinate grid.



Perform the sequence of transformations on $\triangle KRB$. Assume all rotations are centered at the origin. Label the result of the transformations as $\triangle TLS$.

- $(x, y) \rightarrow (x - 12, y + 3)$
- $(x, y) \rightarrow (x, -y)$

Unit 4, Lesson 6: Isometry



Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



Learning Targets

- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can draw an image given a preimage and a sequence of transformations on a coordinate plane.



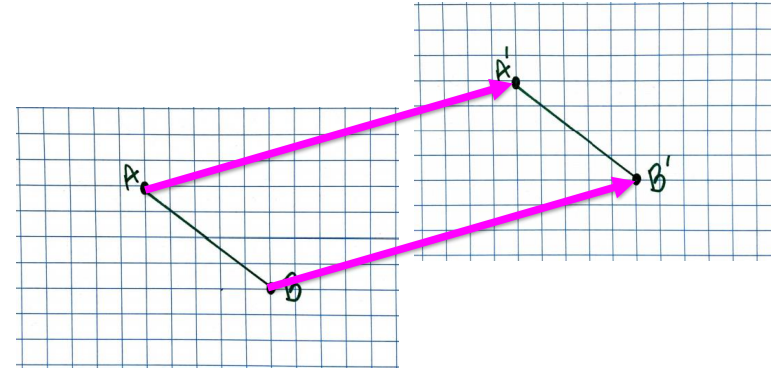
4.6.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Explain your choice.



4.6.2 Exploration: Transforming Parts

1. Traci drew $\overline{AB}$ on a piece of graph paper. She then moved the graph paper up and over to the right, as shown by the arrows. The movement created $\overline{A'B'}$.



The graph paper is a real-life example of a plane. When an object is moved, flipped, or rotated, it is the plane that the object is on that moved. This means that every point on the plane is moved, flipped, or rotated.

- Draw $\overline{YZ}$ on the same plane as $\overline{AB}$. Then, draw $\overline{Y'Z'}$ on the second plane to show how $\overline{YZ}$ would move in the same way as $\overline{AB}$ when the plane is moved.
- Draw the arrows from the endpoints of $\overline{YZ}$ to the endpoints of $\overline{Y'Z'}$. Compare the arrows from $\overline{YZ}$ to $\overline{Y'Z'}$ to the arrows from $\overline{AB}$ to $\overline{A'B'}$. What do you notice?

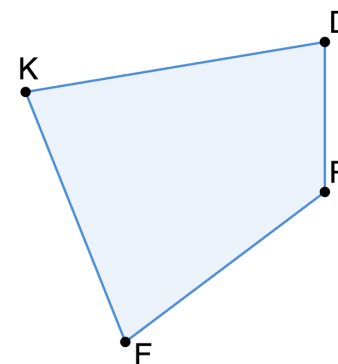
Isometry is a transformation that preserves distance and angle measure.

- c. Use the definition of isometry to justify that Traci's example is an isometry.



4.6.3 Exploration: Transformations and Isometry

1. Quadrilateral $KDRF$ is shown.



- a. Use patty paper to copy the length of $\overline{RF}$. Label the endpoints on the patty paper. Flip the patty paper and place it in the space to the left of quadrilateral $KDRF$. Use the patty paper and any other tools to draw the copy of $\overline{RF}$. Name it $\overline{R'F'}$.

A **transformation** is a one-to-one correspondence between two sets of points.

- b. Work with your partner to complete the statement:

Since there are a set of points on $\overline{RF}$ and a set of points on $\overline{R'F'}$, it is not enough to just state that the endpoints R and F correspond with endpoints R' and F' because a transformation is a _____ between two sets of points. Therefore, to show that $\overline{RF} \cong$ _____, all of the points on $\overline{RF}$ have to have a matching point on $\overline{R'F'}$.

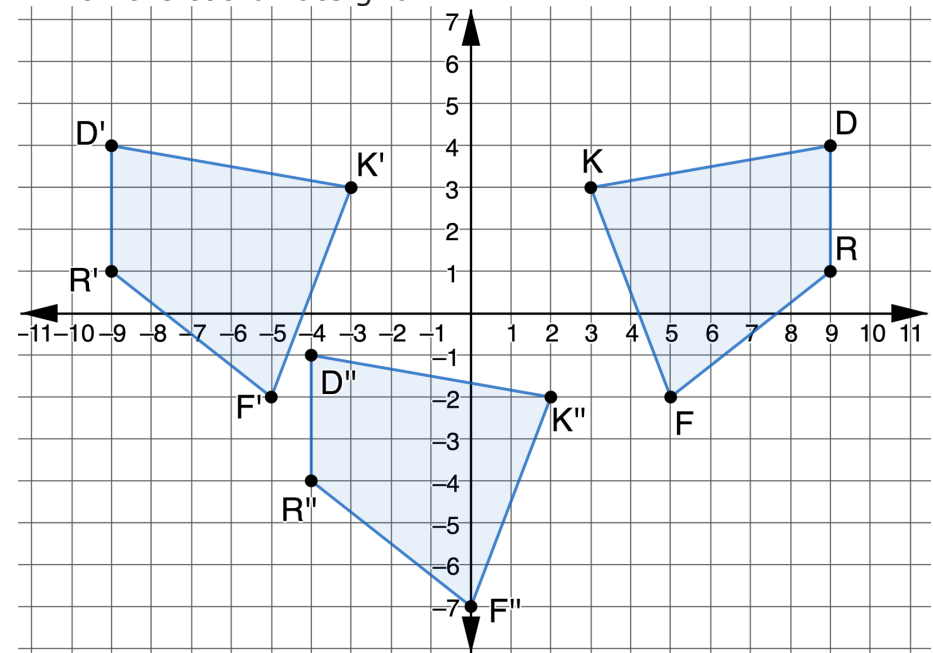


- c. Use the steps to apply the same transformations to each part of quadrilateral $KDRF$.

- Use patty paper to copy $\angle RFK$. Use the copy of $\angle RFK$ to create $\angle R'F'K'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KF}$. Name it $\overline{K'F'}$.
- Use patty paper to copy $\angle FRD$. Use the copy of $\angle FRD$ to create $\angle F'R'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{RD}$. Name it $\overline{R'D'}$.
- Use patty paper to copy $\angle FKD$. Use the copy of $\angle FKD$ to create $\angle F'K'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KD}$. Name it $\overline{K'D'}$.

- d. Explain why when drawing all of the parts of the quadrilateral $KDRF$ to create quadrilateral $K'D'R'F'$, the movements of the patty paper were the same as when drawing $\overline{R'F'}$.

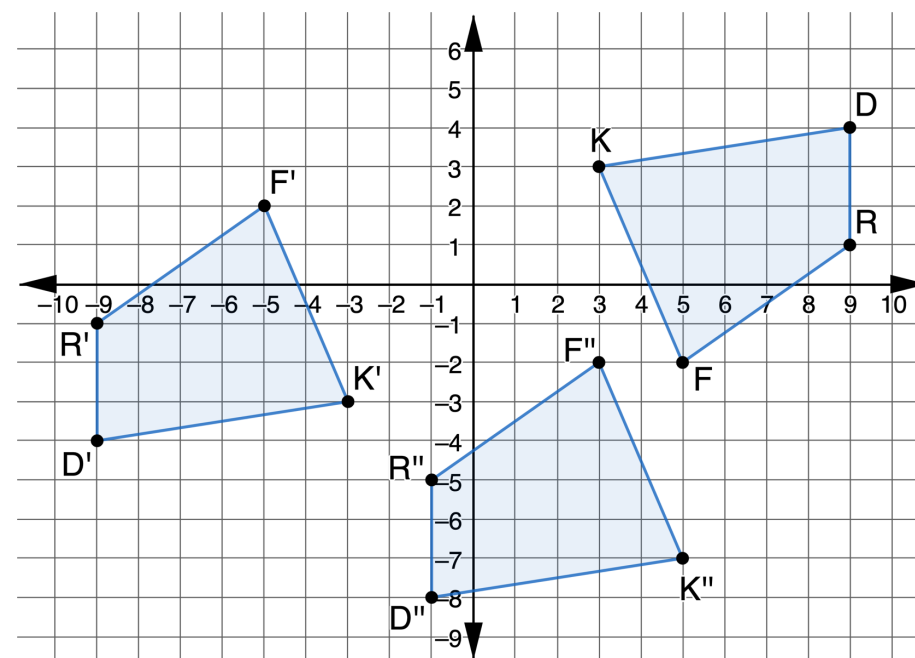
2. Quadrilaterals $KDRF$, $K'D'R'F'$, and $K''D''R''F''$ are shown on the coordinate grid.



- Use patty paper to copy quadrilateral $KDRF$.
- Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $KDRF$.
- Describe what should be done so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K'D'R'F'$.
- Was the movement of the patty paper an example of isometry?

- e. Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $K'D'R'F'$.
 - f. Describe what should be done next so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$.
 - g. Was the movement of the patty paper an example of isometry?
 - h. Determine if the copy of quadrilateral $KDRF$ could be moved with just one transformation so that it lays on top of quadrilateral $K''D''R''F''$.
3. There is at least one other sequence of transformations that can be used to move the copy of quadrilateral $KDRF$ so that it lays on top of quadrilateral $K''D''R''F''$. Work with your partner to describe one of these sequences of transformations.

4. The coordinate grid shows a different sequence of transformations of quadrilateral $KDRF$.



Describe what movements should be done with your patty paper copy of quadrilateral $KDRF$ so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$, by first laying on top of quadrilateral $K'D'R'F'$.



4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

1. For both sequences of transformations of quadrilateral

$KDRF$, the paper copy of quadrilateral $KDRF$ was

- ☐ congruent
- ☐ not congruent

to

- ☐ $K'D'R'F'$ only.
- ☐ $K''D''R''F''$ only.
- ☐ both $K'D'R'F'$ and $K''D''R''F''$.

The transformations used in both sequences were

- ☐ dilations.
- ☐ rigid motions.
- ☐ dilations and rigid motions.

All of the points of

- ☐ $K'D'R'F'$
- ☐ $K''D''R''F''$
- ☐ $K'D'R'F'$ and $K''D''R''F''$

can be mapped to all

of the points of $KDRF$.

2. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $KDRF$ resulted in $KDRF \cong K''D''R''F''$.



Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.



4.6.5 Wrap-Up: Growth Mindset

The power of “YET.”

“The power of believing that you can improve.” - Carol Dweck

Select all of the concepts that apply to your understanding right now.

1. I don’t understand

- isometry
- transformations
- the definition of congruence in terms of rigid motions

YET.

2.

- Isometry
- Mapping
- One-to-one correspondence
- Definition of congruence in terms of rigid motions

doesn’t

make sense to me YET.

3. I can explain

- isometry.
- mapping.
- one-to-one correspondence.
- transformations.
- the definition of congruence in terms of rigid motions.

Unit 4, Lesson 7: Justifying Congruence



Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



Learning Targets

- ☐ I can describe a sequence of transformations that will map a preimage and an image to show congruence.
- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.



4.7.1 Warm-Up: You Can Do It!

"You should be far more concerned with your current trajectory than with your current results." - James Clear

Has anyone ever asked you, "What goals do YOU want to achieve in your lifetime?"



Setting goals is linked to self-confidence, motivation, and autonomy. A 2015 study showed people who wrote down their goals were 33 percent more successful in achieving them than those who just thought about the outcomes. Making goals can be fun.

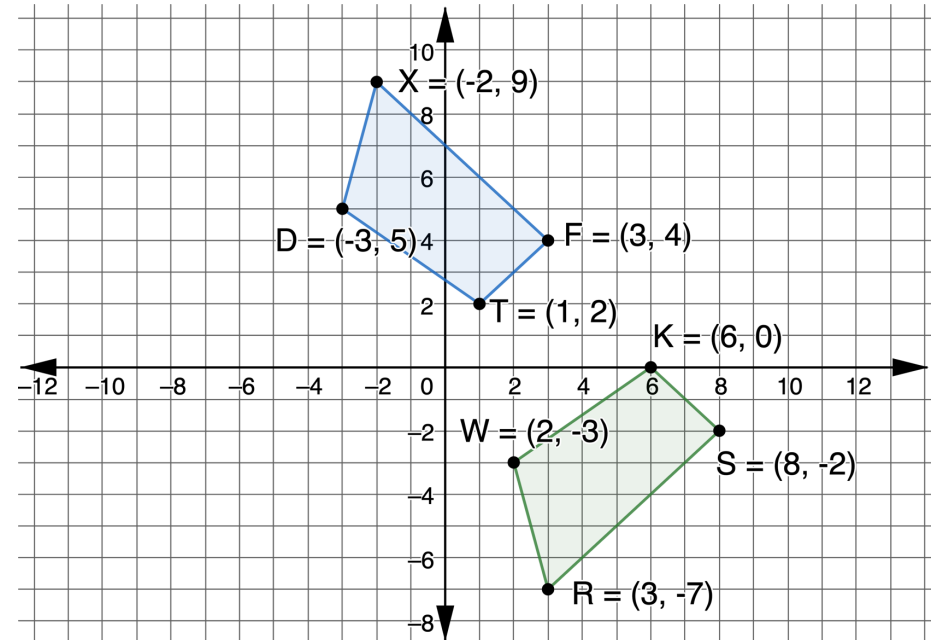
For example, you can name places you would like to visit, activities you would like to do, or people you would like to meet. By setting the goal, you have already set your trajectory toward achieving it. In the table, write three life goals and two goals for the rest of this school year.

Life Goals	Goals for this school year
"If you are always trying to be normal, you will never know how amazing you can be." Maya Angelou	



4.7.2 Guided Instruction: Justifying Congruence

1. Quadrilaterals $XFTD$ and $KSRW$ are shown on the coordinate plane.



- a. Describe a sequence of transformations that will map quadrilateral $KSRW$ onto quadrilateral $XFTD$.

- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $KSRW$ was mapped onto quadrilateral $XFTD$.

Quadrilateral $KSRW$	Quadrilateral $XFTD$
K	
S	
R	
W	



Definition of Congruence in Terms of Rigid Motions

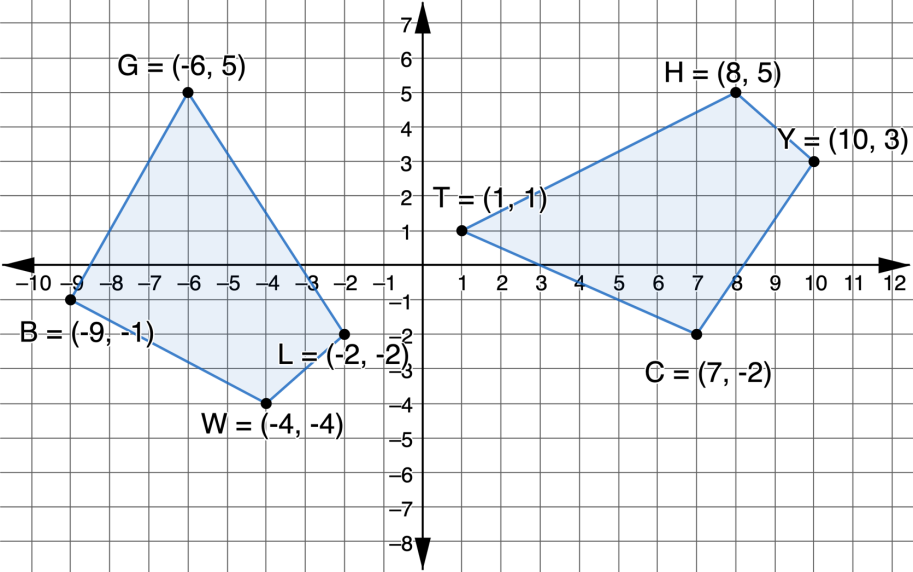
Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

- c. Complete the statements.

Quadrilateral $KSRW$ is congruent to quadrilateral _____ because the sequence of transformations that can be used to map the quadrilaterals were _____.

Because vertex K maps to vertex _____, vertex S maps to vertex _____, vertex R maps to vertex _____, and vertex W maps to vertex _____. The definition of congruence in terms of rigid motions justifies that $KSRW \cong$ _____.

2. Quadrilaterals $BGLW$ and $HYCT$ are shown on the coordinate plane.



- a. Use algebraic descriptions to describe a sequence of transformations that will map quadrilateral $BGLW$ onto quadrilateral $HYCT$.
- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $BGLW$ was mapped onto quadrilateral $HYCT$.

Quadrilateral $BGLW$	Quadrilateral $HYCT$
B	
G	
L	
W	



When using rigid motions to map one figure onto the other, always be sure the corresponding vertices match before stating congruence.

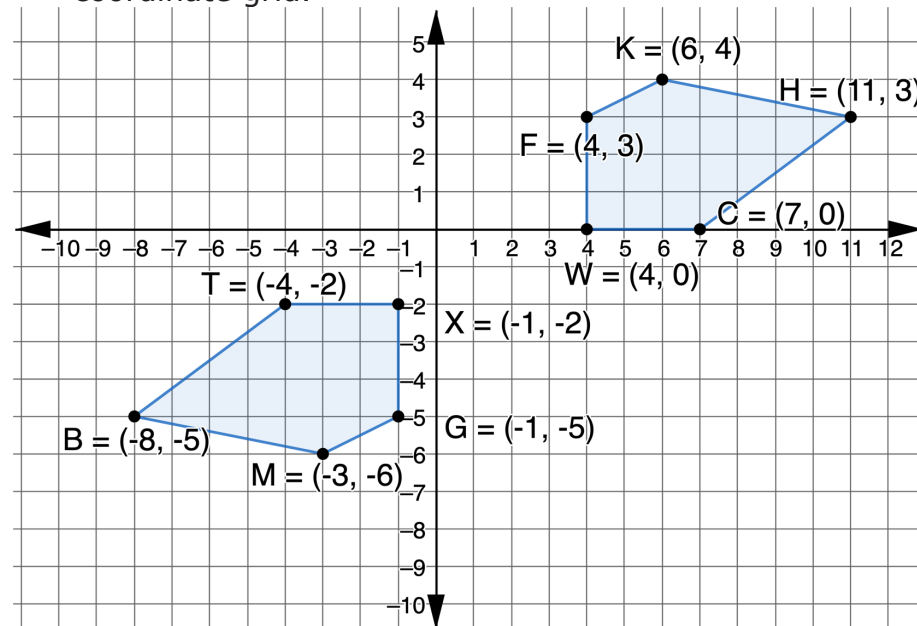
- c. Complete the congruence statement.

$BGLW \cong$ _____



4.7.3 Guided Practice: Justifying Congruence

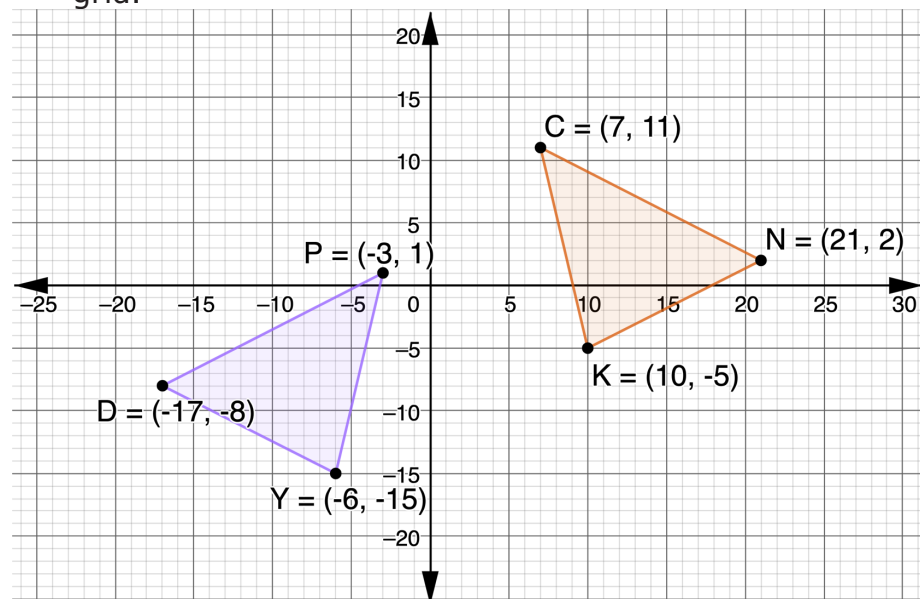
1. Polygons $TXGMB$ and $WFKHC$ are shown on the coordinate grid.



- a. Use algebraic descriptions to describe a sequence of transformations that will map polygon $TXGMB$ onto polygon $WFKHC$.
- b. Complete the congruence statement.

$TXGMB \cong$ _____

2. Triangles CKN and PYD are shown on the coordinate grid.



- a. The sequence of transformations that will map

$\triangle CKN$ onto $\triangle PYD$ is

- translating $\triangle CKN$ left 14,
- reflecting $\triangle CKN$ across the y -axis,
- rotating $\triangle CKN$ 270° clockwise about the origin,

then

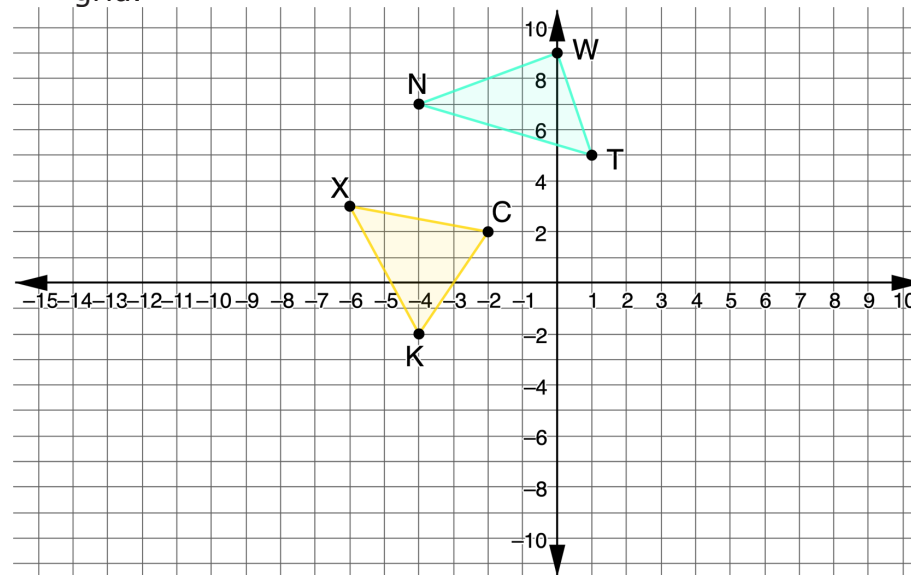
- translating $\triangle C'K'N'$ right 4, down 10.
- translating $\triangle C'K'N'$ right 10, down 4.
- reflecting $\triangle C'K'N'$ across the x -axis.

- b. Vertex C maps to vertex _____, vertex K maps to vertex _____, and vertex N maps to vertex _____ therefore $\triangle CKN$ is congruent to $\triangle$ _____ by the definition of congruence in terms of rigid motions.



4.7.4 Your Turn!

1. Triangles XCK and TWN are shown on the coordinate grid.



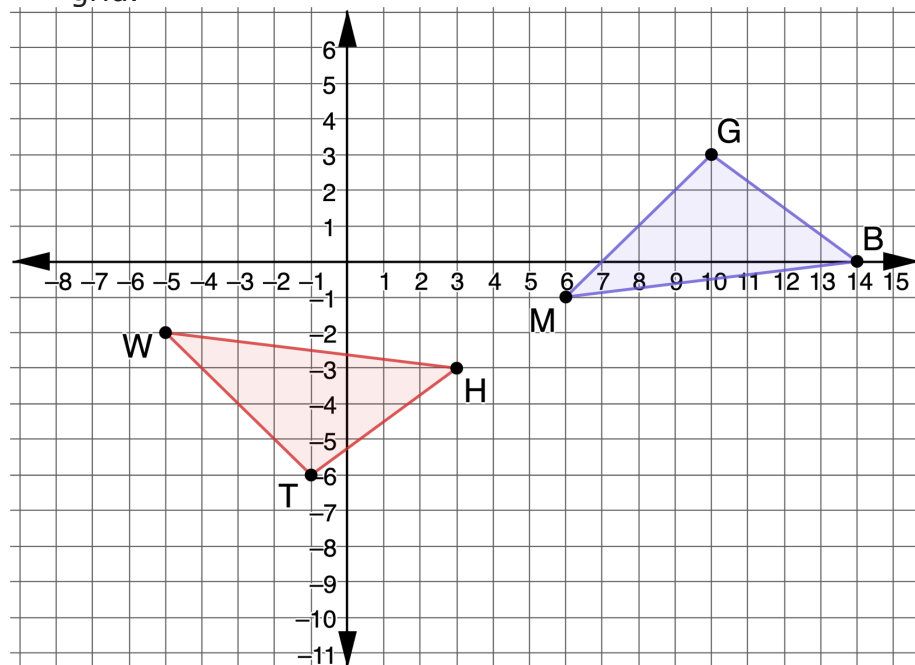
Use algebraic descriptions to describe a sequence of three transformations that will map $\triangle XCK$ onto $\triangle TWN$.

Transformations
$(x, y) \rightarrow$ _____
$(x, y) \rightarrow$ _____
$(x, y) \rightarrow$ _____



4.7.5 Wrap-Up: Ticket Out the Door

1. Triangles TWH and GBM are shown on the coordinate grid.



- a. Use written descriptions or algebraic descriptions to describe a sequence of two transformations that will map the two triangles onto each other.
- b. Complete the table that shows which vertices were mapped to each other.

$\triangle TWH$	$\triangle GBM$
T	
W	
H	

- c. Complete the congruence statement.

$$\triangle TWH \cong \underline{\hspace{2cm}}$$



Unit 4, Lesson 8: Putting It All Together



Benchmarks

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



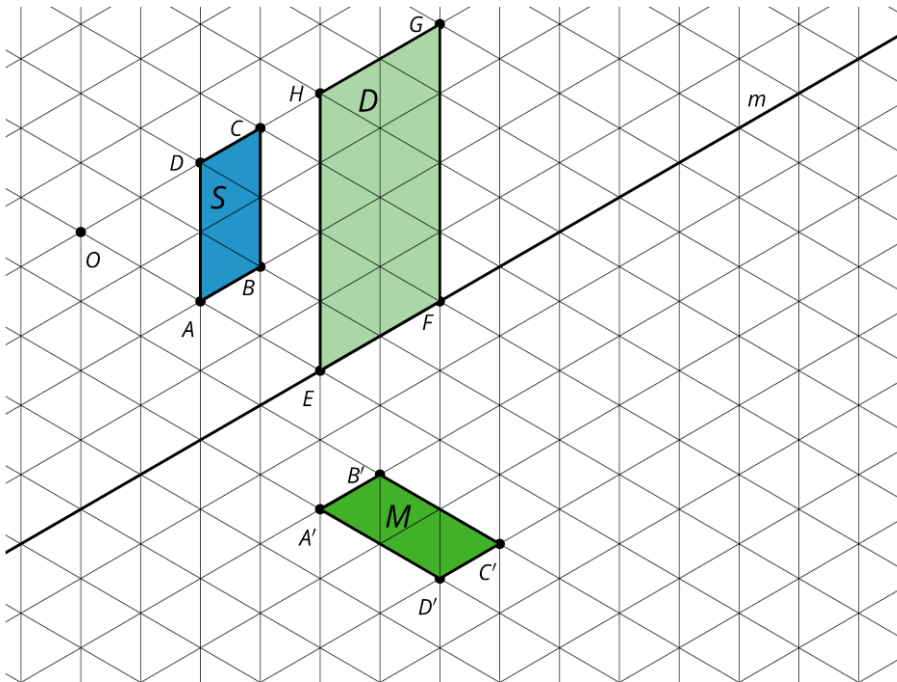
Learning Targets

- ☐ I can determine a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can match the written and algebraic descriptions of a sequence of transformations given the pre-image and image of a figure.



4.8.1 Warm-Up: Notice and Wonder

1. What are three things you notice about the diagram?
2. Reflect quadrilateral $EFGH$ across line m , then translate it up 3 units. Name the resulting quadrilateral $E'F'G'H'$.
3. Check your location of quadrilateral $E'F'G'H'$ with your partner. If your locations differ, explain why both could be correct. If your locations are the same, find another possible interpretation of up 3 units and draw that quadrilateral.



4.8.2 Activity: Card Sort

Match each coordinate grid with the pre-image of $\triangle TLW$ and image $\triangle HPY$ where $\triangle TLW \cong \triangle HPY$, with the written and algebraic descriptions that map $\triangle TLW$ onto $\triangle HPY$. Use the letters in the blue, green, and orange boxes to record your matches in the table.

Coordinate Grid	Written Description	Algebraic Description
B		
D		
F		
G		
H		
M		
T		
W		
X		
Y		



4.8.3 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each target.

1. I can match the written descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



2. I can match the algebraic descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



3. I can determine a sequence of transformations that map one figure onto another congruent figure.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



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